

SQL QUERIES AND THEIR OUTPUTS

1) Attack Type Distribution

```
mysql> select label as attack_type,count(*) as total_attacks from logs where label<>"BENIGN" group by label;
```

attack_type	total_attacks
Web Attack Â? Brute Force	1507
Web Attack Â? XSS	652
Web Attack Â? Sql Injection	21

3 rows in set (0.12 sec)

INSIGHT 1:-

BRUTE FORCE attacks account for most malicious activity

2) Source IP Analysis

```
mysql> select source_ip,count(*) as total_flows from logs group by source_ip order by total_flows desc limit 5;
```

source_ip	total_flows
192.168.10.3	16697
192.168.10.15	16564
192.168.10.17	8936
192.168.10.14	8447
192.168.10.19	8042

5 rows in set (0.58 sec)

INSIGHT 2:-

These are the HIGHEST TRAFFIC SOURCE IP'S (should be investigated)

3) Destination IP Analysis

```
mysql> select destination_ip,count(*) as total_hits from logs group by destination_ip order by total_hits desc limit 5;
```

destination_ip	total_hits
192.168.10.3	37689
192.168.10.1	16374
192.168.10.15	3361
192.168.10.50	3104
172.16.0.1	1937

5 rows in set (0.30 sec)

INSIGHT 3:-

The large traffic difference suggests attack concentration.

4)Port Analysis

```
mysql> select destination_port,count(*) as connection_count from logs group by destination_port order by connection_count desc limit 5;
```

destination_port	connection_count
53	52504
443	22763
80	14934
123	2982
22	513

```
5 rows in set (0.14 sec)
```

INSIGHT 4:-

Port 53 (DNS) is the most frequently used port.

Port 443 (HTTPS) shows extensive encrypted web traffic.

5)Protocol Analysis

```
mysql> select protocol,count(*) as protocol_count from logs group by protocol order by protocol_count ;
```

protocol	protocol_count
0	84
17	56473
6	58605

```
3 rows in set (0.14 sec)
```

INSIGHT 5:-

6(TCP) , 17(UDP) are almost equally used regularly

6)Flow Duration Analysis

```
mysql> select label,round(avg(flow_duration),2) as avg_flow_duration from logs group by label order by avg_flow_duration ;
```

label	avg_flow_duration
Web Attack Â? Sql Injection	2870398.29
Web Attack Â? Brute Force	6506181.63
Web Attack Â? XSS	6715357.09
BENIGN	12654644.30

```
4 rows in set (0.32 sec)
```

INSIGHT 6:-

XSS attacks have the longest average flow duration in label otherthan BENIGN

7) High Risk Score Identification

```
mysql> select source_ip, sum(syn_flag_count+ rst_flag_count + ack_flag_count) as risk_score from logs group by source_ip order by risk_score desc limit 10;
```

source_ip	risk_score
192.168.10.15	2814
192.168.10.50	2528
192.168.10.14	1484
192.168.10.17	1408
192.168.10.8	1248
192.168.10.9	1227
192.168.10.12	1223
192.168.10.5	1205
192.168.10.19	1139
192.168.10.16	990

10 rows in set (0.34 sec)

INSIGHT 7:-

SOURCE IP's with highest risk